Lightweight Object Detection for Driver Monitoring: A Subject-Disjoint Benchmark

Oumar Mamoun Ibrahim
Department of Computer Engineering
University of Sharjah
Sharjah, United Arab Emirates
U22200741@sharjah.ac.ae

Mohamad Khairi bin Ishak
Department of Computer Engineering
University of Sharjah
Sharjah, United Arab Emirates
mishak@sharjah.ac.ae

Khalid Ammar
Department of Electrical and Computer Engineering
Ajman University
Ajman, United Arab Emirates
k.ammar@ajman.ac.ae

Nada Masood Mirza
College of Engineering, Engineering Requirements Unit
United Arab Emirates University
Al Ain, United Arab Emirates
nada.mirza@uaeu.ac.ae

***Abstract*—Reliable driver monitoring requires detectors that generalize to unseen drivers while remaining efficient. This study benchmarks three lightweight object detectors — You Only Look Once (YOLO) 11n, YOLO26n, and nano Fine-grained Distribution Refinement (D-FINE-N) — for visible driver-cue localization under a subject-disjoint protocol. A red–green–blue (RGB) benchmark of 15,723 Driver Monitoring Dataset frames from 14 drivers, including 12,722 negatives, is evaluated across three paired seeds with validation-only operating-point selection and a protected test set. YOLO11n achieves the highest coarse-IoU accuracy and throughput, D-FINE-N the highest calibrated-threshold detection performance, and YOLO26n the strongest strict-IoU localization. Variation across held-out drivers exceeds seed variation, underscoring the importance of subject-level reporting. A complementary eight-system near-infrared study confirms that greater negative training exposure does not consistently improve performance. Detector selection should therefore reflect deployment-specific trade-offs.**

***Index Terms*—driver monitoring, object detection, subject-disjoint evaluation, near-infrared imaging, real-time inference**

I. Introduction

Road traffic injuries cause approximately 1.19 million deaths annually and substantial economic costs worldwide [1]. Euro NCAP now rewards camera-based driver monitoring [2], accelerating in-cabin driver monitoring system (DMS) deployment. Such systems face identity, illumination, viewpoint, and occlusion shifts. Whole-frame recognition may exploit identity or cabin context without exposing its evidence; detection instead localizes cues and makes false positives, misses, and negative-frame errors interpretable. Single-frame detection also avoids temporal-model latency and complexity when throughput is the focus, but it does *not* infer fatigue, intent, duration, or another temporal state.

DMD provides multimodal attention and alertness recordings [3]; Drive&Act provides synchronized color, NIR, depth, and pose streams for fine-grained activities [4]. Both repeatedly observe each driver in temporally correlated video.

Evaluation is optimistic if neighboring frames or identities cross partitions, negatives are removed, test data determine confidence thresholds, or latency boundaries differ. DMD detection studies [5] and posture-classification benchmarks such as State Farm [6] illustrate this methodological diversity and impede cross-paper ranking.

Recent work proposes lightweight in-cabin recognition and detection networks [7]–[10]. We compare the localization, reliability, false-positive behavior, and computation of complete public detectors on unseen drivers without architectural modification. A separate exploratory NIR experiment tests whether greater negative exposure changes performance consistently across model families. The contributions are:

- a primary subject-disjoint RGB benchmark for four explicitly defined visible cues, with all sampled frames preserved and annotation structure automatically audited;
- validation-only operating-point selection followed by protected testing, three paired RGB seeds, class- and subject-level analysis, false detections on negative frames, and synchronized latency; and
- a checksum-backed workflow and separate one-seed, eight-system NIR negative-exposure study with deterministic 1-FPS midpoint sampling.

II. Related Work

A. Driver Recognition and Cue Localization

MobileVGG and convolution–transformer hybrids provide efficient distracted-driver classification [7], [11], while PO-GUISE+ uses pose and object guidance on Drive&Act NIR imagery [12]. These whole-frame or temporal tasks differ from bounding-box detection, which must identify a cue and its spatial evidence in one frame without estimating temporal state.

Detection work modifies YOLO backbones, attention, or multiscale aggregation for dangerous-driving or fatigue cues

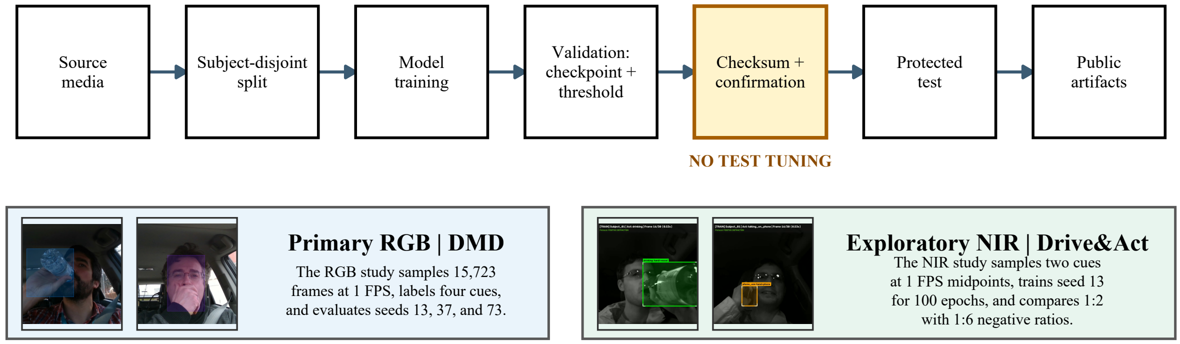

Fig. 1. End-to-end benchmark workflow with representative annotated DMD and Drive&Act frames. RGB and NIR share the validation-before-test gate but retain dataset-specific sampling, ontology, seeds, and training conditions. FPS denotes frames per second.

[8]–[10]. Comparisons cover convolutional networks versus transformers [13] and multiple YOLO generations [14]. YOLO evolved from single-shot grid regression [15] through successive revisions [16]–[18], but driver separation, retained negatives, calibration, and runtime protocols differ across studies.

B. Released Detector Systems

The Detection Transformer (DETR) uses bipartite set prediction [19]; RT-DETR enables real-time use [20]; RT-DETRv2 adds a bag-of-freebies baseline [21]; and D-FINE refines box-edge distributions [22]. Both tracks use shipped D-FINE-N, YOLO11n [17], and YOLO26n [18]; NIR adds YOLOv8n [23], YOLOX-Nano [24], YOLOv10n [25], SSDLite-MobileNetV3-Large [26], and RT-DETRv2-S [21]. Native recipes remain part of each system: YOLO26 and D-FINE are end-to-end, whereas YOLO11n retains non-maximum suppression (NMS) [27]. Lightweight systems can approach real-time performance on low-cost hardware [28]; this is therefore a system, not architecture-only, benchmark.

No prior comparison of shipped lightweight detectors combines subject-disjoint splitting, validation-only thresholds, a protected test set, all negative frames, and synchronized latency; this benchmark does.

III. Methodology

A. Task Definition and Workflow

The task localizes four visible RGB cues: yawning, hand over mouth, drinking, and phone use. Bounding boxes expose spatial evidence and negative-frame errors while discouraging identity or cabin shortcuts. Figure 1 shows construction, splitting, training, and validation before a checksum-confirmation gate; no test-derived threshold or model change is allowed. RGB is primary, and NIR is a distinct secondary experiment rather than a paired modality comparison.

B. RGB Dataset and Annotation Protocol

The RGB set contains 15,723 face crops sampled at 1 FPS from 81 DMD recordings of 14 drivers. One annotator completed one pass, without an independent semantic-agreement review. All frames remain without post-sampling curation: 3,001 positive and 12,722 negative. Table I fixes the box ontology.

TABLE I
Frozen RGB Box Semantics and Test Support

Cue	Bounding-box target	Test n	Median area
Yawning	Mouth	33	1.38%
Hand over mouth	Full face and covering hand	28	24.67%
Drinking	Visible hand/container–mouth interaction	56	16.09%
Phone use	Visible hand–phone interaction	497	8.37%

Each frame has zero or one box. Yawning and hand-over-mouth are exclusive; the latter takes precedence when a hand covers a yawn. Boxes contain only visible targets: unobserved anatomy is not estimated, but identifiable partial occlusion remains valid. No minimum-pixel cutoff is imposed at 640×640.

An automated COCO audit found 15,723 unique images and 3,001 annotations, with no duplicate IDs or file names, orphans, invalid categories, nonfinite, nonpositive, or out-of-bounds boxes, or multi-annotation frames. The smallest box occupied 0.81% of a frame; 139/159 yawning boxes were below 2%. This verifies structure, not semantic agreement or label accuracy.

C. Split, Models, and Training

An exhaustive 8:3:3 assignment minimizes worst relative class-distribution deviation, breaking ties by root-mean-square deviation. Frozen training, validation, and test sets contain 9,087, 3,423, and 3,213 frames. Test subjects contribute 1,080, 920, and 1,213 frames, including 178, 183, and 253 positives.

COCO-pretrained YOLO11n [17], YOLO26n [18], and D-FINE-N [22] use 640×640 input, physical/effective batches of 8/32 via four-step accumulation, 220 epochs, automatic mixed precision (AMP), and no early stopping. A fixed patch normalizes incomplete final accumulation by its actual sample count. Native recipes are Ultralytics `optimizer=auto` with a linear schedule and D-FINE AdamW/MultiStepLR. Seeds

TABLE II
SHARED RGB TRAINING CONFIGURATION

Setting	Value
Input resolution	640×640
Physical / effective batch	8 / 32 (4-step accumulation)
Epochs	220 (no early stopping)
Precision	AMP (FP16/FP32)
Pretrained backbone	COCO
Seeds	$\{13, 37, 73\}$ (predeclared)
YOLO optimizer	`auto` + linear schedule
D-FINE optimizer	AdamW + MultiStepLR

$\{13, 37, 73\}$ were predeclared; no run was selected for presentation. Table II summarizes settings.

D. Validation, Test Metrics, and Profiling

The best native validation checkpoint is retained. Threshold τ^* maximizes validation micro-F_1 on $[0.01, 0.99]$, breaking ties by precision then τ. At intersection over union (IoU) 0.5, predictions match greedily one-to-one within class. Protected tests report COCO mAP_{50} and $\mathrm{mAP}_{50:95}$ [29], precision, recall, micro- and macro-F_1, and

$$\mathrm{FD}_{100} = 100\frac{\mathrm{FP}_{\mathrm{neg}}}{N_{\mathrm{neg}}}, \tag{1}$$

where $\mathrm{FP}_{\mathrm{neg}}$ counts above-threshold detections on negative frames. This diagnostic differs from false discovery rate $\mathrm{FP}/(\mathrm{FP}+\mathrm{TP})$: it measures firing where no target cue exists. It is a static frame count, not a nuisance-alert model; no temporal confirmation or debouncing is applied.

RGB results give sample mean and standard deviation (SD) over three seeds. With a fixed subject split, SD measures optimization variability, not uncertainty across drivers. We therefore report each test subject and check paired differences for consistent signs; no significance test is claimed for $n = 3$.

All model selection and operating-point determination use validation data exclusively; the fixed test partition is used only for final evaluation.

Batch-1 FP16 profiling uses an NVIDIA RTX 4060 8-GB GPU. CUDA synchronization brackets model-forward and tensor-to-final intervals; disk I/O, decoding, preprocessing, and metrics are excluded. We report p50 latency and sustained FPS, not CPU or embedded performance.

E. Exploratory NIR Protocol

The separate NIR experiment asks whether more training negatives consistently reduce false detections across a broader model set. Its different dataset makes it complementary to, not comparable with, the single-condition RGB track.

The NIR pool has 3,786 one-second front-top active-NIR (850 nm) Drive&Act snippets from 15 drivers; excluding 23 unilluminated tasks leaves 3,763. Drinking and phone-use tracklets define two positive classes. Each snippet yields its deterministic 1-FPS midpoint ($t = 0.5$ s; zero-based offset 14). Normalized box coordinates (x, y, width, height) are linearly interpolated between the nearest bracketing human-labeled keyframes, or taken from the nearest endpoint when no pair brackets the midpoint. Driver crops are resized to 640×640. Thus NIR applies dataset-specific exclusion and derives midpoint boxes from human tracklets.

A fixed 9:3:3 subject split serves all 16 model–condition runs across one-stage convolutional, SSD, and transformer families. Validation and test examples are identically ordered. Training contains 270 positives and 540 or 1,620 negatives (ratios 1:2 and 1:6); the 1:2 set is a subject-stratified seed-13 subset of the 1:6 pool.

All runs use batch 8, four-step accumulation, 100 epochs without early stopping, the best native validation checkpoint, and native optimizers/schedules; D-FINE changes augmentation at epoch 67. Ratios 1:2 and 1:6 yield 2,600 and 6,000 updates. The latter has three times the unique negatives and 2.31 times the updates, making this a *training-negative exposure study*, not a matched-signal causal ratio ablation.

All NIR jobs use seed 13 and RGB operating metrics. One frame represents each snippet, so no temporal aggregation occurs. Micro-F_1 and FD_{100} remain detection-level; stored macro-F_1 is class-presence localization, counting repeated same-class detections once per frame.

IV. RESULTS AND ANALYSIS

A. Aggregate RGB Results

Across nine RGB runs (Table III), YOLO11n leads mAP_{50} and throughput, YOLO26n leads $\mathrm{mAP}_{50:95}$, and D-FINE-N leads both F_1 measures. D-FINE-N's calibrated-point advantage costs 8.0–26.4 FPS and does not extend to strict-IoU AP; the result is a trade-off, not universal superiority.

B. Class Imbalance and Paired-Seed Evidence

Phone use supplies 497/614 positive tests, versus 28 hand-over-mouth, 33 yawning, and 56 drinking instances. This can affect representations, validation thresholds, and pooled micro-F_1, but not directly dominate COCO mAP because category APs receive equal weight.

YOLO26n minus YOLO11n class AP is $+6.21$ points for hand over mouth, -3.85 for yawning, $+1.02$ for drinking, and $+3.02$ for phone use. D-FINE-N leads yawning, drinking, and phone use but trails hand over mouth. Macro-F_1 and per-class AP limit majority-class bias, although wide low-support SDs warrant caution. All three seeds give YOLO26n a consistent $\mathrm{mAP}_{50:95}$ advantage over YOLO11n despite the small seed count.

C. Held-Out Subject Sensitivity

For Subjects 05, 10, and 12, mean micro-F_1 is 94.94%, 62.41%, and 94.72% for YOLO11n; 96.11%, 54.11%, and 92.29% for YOLO26n; and 98.20%, 78.79%, and 94.30% for D-FINE-N. Between-subject variation exceeds pooled seed SD, so pooling hides the difficult driver.

D. Qualitative Error Analysis

Figure 3 shows predeclared seed-13 frozen predictions. Correct detections localize the interaction; false negatives expose glare and small drinking targets. Cue-like nominal negatives may reflect ontology boundaries or missed labels, so structural audits cannot establish semantic validity. Retaining them exposes the principal annotation risk.

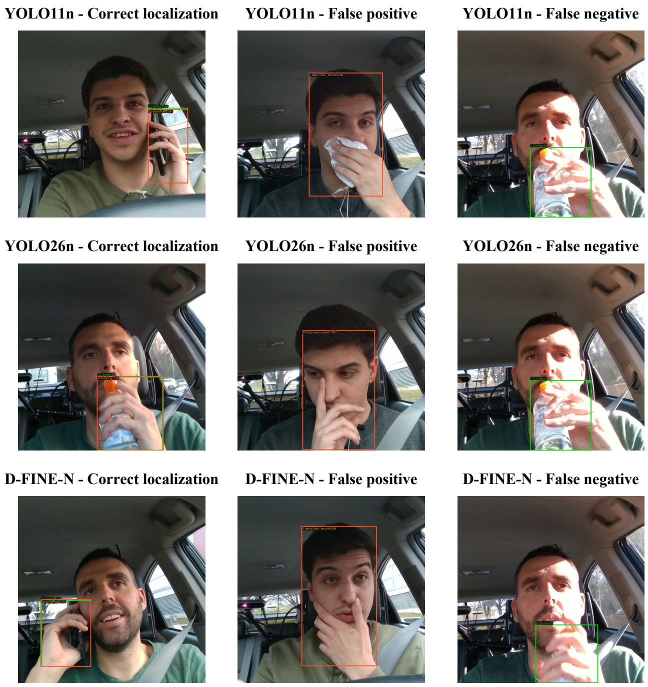

Fig. 3. Representative seed-13 outcomes for YOLO11n, YOLO26n, and D-FINE-N. Green boxes denote ground truth and orange boxes denote predictions at the validation-selected threshold; columns show a correct localization, a false detection on a nominal negative, and a false negative.

E. Exploratory NIR Results

All 16 protected NIR passes are complete. Table IV reports two exposures for eight systems; Fig. 5 plots the same checksum-backed results, and Fig. 4 shows agreement-selected outcomes across runs.

Changing 1:2 to 1:6 improves $\mathrm{mAP}_{50:95}$ only for YOLO26n, SSDLite-MobileNetV3-Large, and RT-DETRv2-S, with changes from $+5.52$ points (YOLO26n) to -7.71 (YOLOX-Nano). Only SSDLite improves micro-F_1. Thus the effect is model-dependent. At both ratios, RT-DETRv2-S leads strict-IoU AP, while YOLOv8n leads micro-F_1 and tensor-to-final p50 latency.

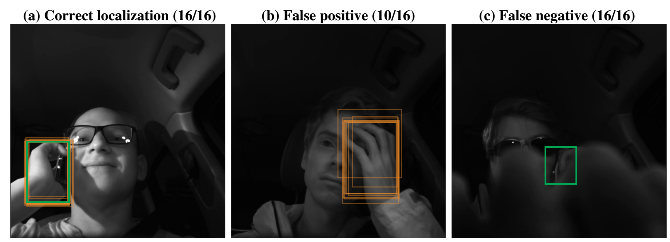

Fig. 4. Agreement-selected NIR outcomes at frozen thresholds. Green denotes ground truth and orange denotes predictions. The correct localization and false negative are unanimous; the most recurrent negative-frame false positive occurs in 10/16 runs.

V. Discussion

A. Performance and Deployment Trade-offs

Results trade localization, reliability, and efficiency. In RGB, YOLO11n balances coarse-IoU accuracy and through-

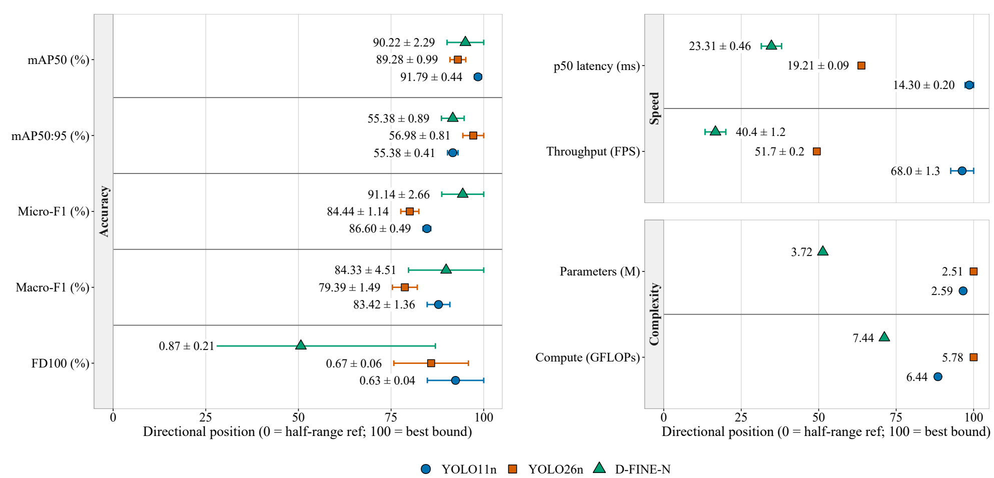

Fig. 2. Directional dot-and-whisker plot for YOLO11n, YOLO26n, and D-FINE-N. Rows are grouped into accuracy (left), speed, and complexity (right); right indicates better performance, and whiskers show sample SD across seeds.

TABLE III

Mean and sample standard deviation across three seeds for the RGB track on the protected test split. Bold underlined values indicate the best result per column.

Model	mAP_{50}	$\mathrm{mAP}_{50:95}$	Micro-F_1	Macro-F_1	FD_{100}	p50 (ms)	FPS	Params (M)	GFLOPs
YOLO11n	**<u>91.79±0.44</u>**	55.38±0.41	86.60±0.49	83.42±1.36	**<u>0.63±0.04</u>**	**<u>14.30±0.20</u>**	**<u>68.0±1.3</u>**	2.59	6.44
YOLO26n	89.28±0.99	**<u>56.98±0.81</u>**	84.44±1.14	79.39±1.49	0.67±0.06	19.21±0.09	51.7±0.2	**<u>2.51</u>**	**<u>5.78</u>**
D-FINE-N	90.22±2.29	55.38±0.89	**<u>91.14±2.66</u>**	**<u>84.33±4.51</u>**	0.87±0.21	23.31±0.46	40.4±1.2	3.72	7.44

TABLE IV
EXPLORATORY NIR TEST RESULTS AT SEED 13. PER-MODEL RATIO WINNERS ARE BOLD UNDERLINED; DISPLAYED TIES ARE UNMARKED.

Model	Neg.	$\mathrm{mAP}_{50:95}$	Micro-F_1	Macro-F_1	FD_{100}	p50
YOLO11n	1:2	**42.62**	**71.91**	61.94	**0.14**	15.22
YOLO11n	1:6	37.07	69.15	**70.32**	0.41	**13.74**
YOLO26n	1:2	35.75	**76.54**	**83.62**	2.17	17.11
YOLO26n	1:6	**41.28**	65.96	65.57	**0.68**	**16.84**
D-FINE-N	1:2	**46.38**	**83.58**	**80.25**	**0.54**	24.58
D-FINE-N	1:6	46.30	79.23	76.45	0.81	**24.10**
SSDLite-MNV3-L	1:2	28.43	73.39	69.97	3.11	**14.76**
SSDLite-MNV3-L	1:6	**30.76**	**78.73**	**73.38**	**1.35**	15.93
RT-DETRv2-S	1:2	48.14	**84.42**	**78.14**	0.27	25.61
RT-DETRv2-S	1:6	**48.51**	79.79	76.80	0.27	**24.66**
YOLOX-Nano	1:2	**33.95**	**79.23**	**76.36**	0.95	**15.46**
YOLOX-Nano	1:6	26.24	67.37	61.31	**0.41**	16.24
YOLOv10n	1:2	**36.67**	**69.23**	**65.40**	2.17	15.41
YOLOv10n	1:6	32.58	66.67	64.32	**2.03**	**13.45**
YOLOv8n	1:2	**39.21**	**84.51**	**83.71**	0.27	11.74
YOLOv8n	1:6	37.77	80.81	76.75	0.27	**10.24**

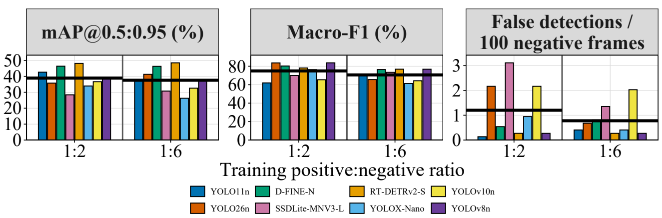

Fig. 5. Sixteen NIR protected-test results at seed 13 and deterministic 1-FPS midpoint sampling. Distinct colors identify the eight models at 1:2 and 1:6 training exposures; the dark-grey rule separates the ratio sections, and each thick black segment marks the eight-model mean for its ratio.

put, YOLO26n leads strict-IoU localization, and slower D-FINE-N leads F_1. NIR likewise varies by model and criterion. No detector satisfies every requirement; selection depends on deployment accuracy, reliability, and efficiency needs.

B. Class and Subject Effects

The phone-use majority limits pooled metrics but does not invalidate YOLO26n's strict-IoU advantage: equal-weight AP, macro-F_1, supports, and paired-seed directions separate it from majority dominance. The narrow margin, three low-support cues, and drinking sign reversal support a trade-off rather than a rank. Subject 10 shows that single-split seed variation understates variation across people.

C. Interpretation Scope

Subject-disjoint evaluation reduces identity leakage but 14 drivers and three held-out identities cannot establish population generalization. One annotator and one pass provide consistency but no independent semantic agreement; the nominal-negative example illustrates this risk. Future rounds should estimate agreement (e.g., Cohen's κ on stratified cue subsets). RGB retains within-recording correlation, and single frames neither estimate fatigue nor model alert debouncing.

NIR's midpoint avoids within-snippet duplication but 1 FPS omits intra-second motion. NIR also differs in dataset, ontology, annotations, preprocessing, and seed count, while changing negative diversity with training steps; it is not a controlled RGB-versus-NIR effect.

RTX 4060 timing excludes preprocessing and embedded hardware. These boundaries preserve the benchmark's value as a transparent, reproducible comparison.

VI. CONCLUSION AND FUTURE WORK

This subject-disjoint RGB benchmark and complementary NIR negative-exposure experiment show that no detector dominates: coarse-IoU accuracy and throughput, strict-IoU localization, and calibrated detection reliability favor different systems.

If false alarms are costlier than misses, YOLO11n's lower FD_{100} and highest throughput are attractive. If downstream gaze or posture needs precise localization, YOLO26n's strict-IoU advantage matters. If latency can buy calibrated F_1, D-FINE-N offers the highest reliability. Subject 10's large gap shows that aggregates can hide deployment-critical failures; subject-level reporting should be standard.

Future work should expand held-out-driver number and diversity, add independent annotation review, repeat multiple subject partitions, and evaluate complete pipelines on embedded hardware.

Data and code: Authored annotations, frozen partitions/checksums, sanitized predictions, evaluation tools, and artifacts are at https://github.com/IamOumarIbrahim/lightweight-driver-behavior-detection; licensed DMD/Drive&Act media and checkpoints remain local under original terms.

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